

RCME

Installation and operating instructions



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Declaration of conformity

GB: EC declaration of conformity

We, Grundfos, declare under our sole responsibility that the product RCME, to which this declaration relates, is in conformity with these Council directives on the approximation of the laws of the EC member states:

- Machinery Directive (2006/42/EC).
Standard used: EN809:2008.
- Low Voltage Directive (2006/95/EC).
Standards used: EN 60204-1:2006 + A1:2009.
- EMC Directive (2004/108/EC).
Standards used: EN 61000-6-2:2005 and EN 61000-6-3:2007.

This EC declaration of conformity is only valid when published as part of the Grundfos installation and operating instructions (publication number 98136614 0412).

DE: EG-Konformitätserklärung

Wir, Grundfos, erklären in alleiniger Verantwortung, dass das Produkt RCME, auf das sich diese Erklärung bezieht, mit den folgenden Richtlinien des Rates zur Angleichung der Rechtsvorschriften der EU-Mitgliedsstaaten übereinstimmt:

- Maschinenrichtlinie (2006/42/EG).
Norm, die verwendet wurde: EN809:2008.
- Niederspannungsrichtlinie (2006/95/EG).
Normen, die verwendet wurden: EN 60204-1:2006 + A1:2009.
- EMV-Richtlinie (2004/108/EG).
Normen, die verwendet wurden: EN 61000-6-2:2005 und EN 61000-6-3:2007.

Diese EG-Konformitätserklärung gilt nur, wenn sie in Verbindung mit der Grundfos Montage- und Betriebsanleitung (Veröffentlichungsnummer 98136614 0412) veröffentlicht wird.

NL: EC overeenkomstigheidsverklaring

Wij, Grundfos, verklaren geheel onder eigen verantwoordelijkheid dat het product RCME waarop deze verklaring betrekking heeft, in overeenstemming is met de Richtlijnen van de Raad in zake de onderlinge aanpassing van de wetgeving van de EG lidstaten betreffende:

- Machine Richtlijn (2006/42/EC).
Gebruikte norm: EN809:2008.
- Laagspannings Richtlijn (2006/95/EC).
Gebruikte normen: EN 60204-1:2006 + A1:2009.
- EMC Richtlijn (2004/108/EC).
Gebruikte normen: EN 61000-6-2:2005 en EN 61000-6-3:2007.

Deze EC overeenkomstigheidsverklaring is alleen geldig wanneer deze gepubliceerd is als onderdeel van de Grundfos installatie- en bedieningsinstructies (publicatienummer 98136614 0412).

DK: EF-overensstemmelseserklæring

Vi, Grundfos, erklærer under ansvar at produktet RCME som denne erklæring omhandler, er i overensstemmelse med disse af Rådets direktiver om indbyrdes tilnærmelse til EF-medlemsstaternes lovgivning:

- Maskindirektivet (2006/42/EF).
Anvendt standard: EN809:2008.
 - Lavspændingsdirektivet (2006/95/EF).
Anvendte standarder: EN 60204-1:2006 + A1:2009.
 - EMC-direktivet (2004/108/EF).
Anvendte standarder: EN 61000-6-2:2005 og EN 61000-6-3:2007.
- Denne EF-overensstemmelseserklæring er kun gyldig når den publiceres som en del af Grundfos-sikkerhedsanvisningerne (publikationsnummer 98136614 0412).

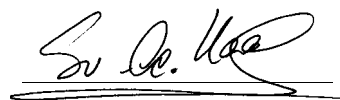
FR : Déclaration de conformité CE

Nous, Grundfos, déclarons sous notre seule responsabilité, que le produit RCME, auquel se réfère cette déclaration, est conforme aux Directives du Conseil concernant le rapprochement des législations des Etats membres CE relatives aux normes énoncées ci-dessous :

- Directive Machines (2006/42/CE).
Norme utilisée: EN809:2008.
- Directive Basse Tension (2006/95/CE).
Normes utilisées: EN 60204-1:2006 + A1:2009.
- Directive Compatibilité Electromagnétique CEM (2004/108/CE).
Normes utilisées: EN 61000-6-2:2005 et EN 61000-6-3:2007.

Cette déclaration de conformité CE est uniquement valide lors de sa publication dans les consignes de sécurité Grundfos (numéro de publication 98136614 0412).

Bjerringbro, 2nd January 2012



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Person authorised to compile technical file and empowered to sign the EC declaration of conformity.

Original installation and operating instructions.

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**Warning**

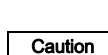
Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.

1. Symbols used in this document**Warning**

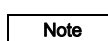
If these safety instructions are not observed, it may result in personal injury.

**Warning**

If these instructions are not observed, it may lead to electric shock with consequent risk of serious personal injury or death.

**Caution**

If these safety instructions are not observed, it may result in malfunction or damage to the equipment.

**Note**

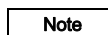
Notes or instructions that make the job easier and ensure safe operation.

2. Scope of these instructions

These installation and operating instructions apply to the Grundfos RCME rainwater pumping system. For instructions not mentioned specifically in this manual, see the installation and operating instructions supplied with the SPO feed pump.

3. Delivery and handling

The Grundfos RCME pressure-boosting unit, including all sub-components, is delivered from factory on a wooden pallet designed for transport by fork-lift truck or a similar vehicle.

**Note**

To ensure safe transport, we recommend that the system is transported with suitable lifting tools.

4. Applications

The Grundfos RCME rainwater system is designed to supply collected rainwater to toilets and laundries in large buildings such as:

- shopping malls
- schools
- office buildings
- hospitals and nursing homes
- green houses and farms
- multi-family homes, housing complexes, etc.

5. Product description

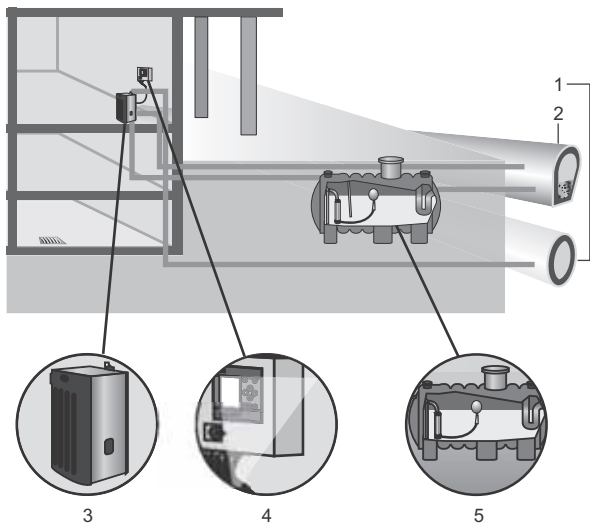


Fig. 1 System principle

Pos.	Description
1	Mains water supply
2	Wastewater
3	RCME pressure-boosting unit
4	Control cabinet with CU 361 control unit
5	Rainwater collecting tank with SPO pump, floating suction strainer and hydrostatic pressure sensor

The Grundfos RCME rainwater system consists of a pressure-boosting unit and a control cabinet with a Grundfos CU 361 control unit. The system also includes a Grundfos SPO submersible pump which functions as feed pump for the RCME pressure-boosting unit. Install the feed pump in a rainwater collecting tank.

The RCME unit incorporates back-up of mains water supply. This function is automatically activated if the collecting tank is emptied, for example during long periods without rain.

The RCME system is operated and monitored via the user-friendly control panel on the CU 361 control unit that monitors and controls the rainwater system.

The RCME system can communicate in many different ways, for example with a building management system.

5.1 Overview of system components

The RCME system consists of the following components:

- RCME 5-4 pressure-boosting unit
 - CME 5-4 booster pump
 - buffer tank
 - integrated pressure vessel
 - mains water valve
 - steel base plate with outlet connection
- control cabinet with CU 361 control unit
- SPO 5-55 submersible feed pump with floating suction strainer
- hydrostatic pressure sensor for collecting tank
- hose for connection between the booster pump and the base plate outlet connection.

6. Identification

6.1 Nameplate

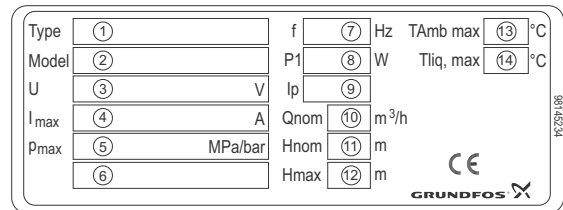


Fig. 2 Nameplate

Pos.	Description
1	Type designation
2	Model
3	Supply voltage [V]
4	Maximum current [A]
5	Maximum pressure [MPa/bar]
6	Country of origin
7	Frequency [Hz]
8	Input power [kW]
9	Enclosure class
10	Rated flow [m³/h]
11	Head at rated flow [m]
12	Maximum head [m]
13	Maximum ambient temperature [°C]
14	Maximum liquid temperature [°C]

6.2 Type key

Example	R	CME	05	03
Rainwater				
Pump type				
CME: Centrifugal Modular booster pump with integrated frequency converter				
Rated flow rate				
Rated flow rate at 50 Hz [m³/h]				
Number of phases of the electrical supply				

7. Mechanical installation



Warning

Installation and operation must comply with local regulations and accepted codes of good practice.

Checks before installation:

- Do the RCME unit, control cabinet and feed pump correspond to order?
- Are all visible parts intact?

7.1 Location

7.1.1 RCME unit

The RCME unit should be sited in a dry, well-ventilated, and frost-free position where it will not be exposed to extremes of temperature. In order to reduce bacteria growth, the ambient temperature should ideally be less than +20 °C. The unit may be located outdoors in a weather-, frost- and rodent-proof enclosure with adequate ventilation, especially during hot weather.

The RCME unit must not be installed into roof spaces. The vibration level associated with any rotating equipment will cause disturbance, and if a leak occurs, considerable water damage could result.

If the RCME unit is installed in a basement and the buffer tank is installed below the collecting tank, siphoning may occur from the collecting tank to the buffer tank. In order to prevent this, an anti-siphoning valve should be installed or the pipework should be carried out in such a way that siphoning is prevented.

Provide drainage facilities for the unit for use in case of leakage, overflow or loss of water during commissioning and service.

Position the RCME unit to allow access for maintenance and service.

Provide adequate drainage facilities and protection from water damage in the immediate vicinity of the RCME unit.

7.1.2 Control cabinet

The control cabinet is designed for wall mounting.

The CU 361 must not be exposed to direct sunlight.

7.1.3 SPO 5-55 feed pump

Install the feed pump according to the installation and operating instructions supplied with the pump.

7.2 Foundation

Position the RCME unit on the supplied base plate. The base plate must be placed on an even and solid surface capable of supporting the weight of the unit with filled buffer tank, for instance on a concrete floor or foundation.

See section 19.7 *Weight of RCME unit*.

7.3 Pipework

Caution

All external pipes must be properly supported before they are connected to the RCME unit to ensure that the internal pipework is not stressed.

Note

We recommend to fit isolating valves on either side of the unit. It is thus not necessary to drain the system if the RCME unit requires service.

All pipework subject to freezing conditions must be adequately insulated.

7.3.1 Rainwater inlet pipe from collecting tank

For thread size and location of the rainwater inlet pipe, see fig. 1, page 102, and fig. 2, page 103.

7.3.2 Mains water supply pipe and flow sensor

Install the flow sensor in the mains water inlet pipe before the RCME unit. Observe the flow indication on the flow sensor pipe/ housing.

For thread size and location of the mains water supply pipe, see fig. 1, page 102, and fig. 2, page 103.

7.3.3 Discharge pipe

Connect the hose supplied with the system between the booster pump and the base plate outlet.

Connect the piping from the building to the hose connection via an isolating valve.

For thread size and location of the mains water supply pipe, see fig. 1, page 102, and fig. 2, page 103.

7.3.4 Overflow

Lead the overflow to a drain via a funnel to ensure that there is an air gap between the buffer tank and the pipe leading to the sewer. See the installation example in fig. 1, page 102.

8. Electrical installation

Warning

The electrical connection must be carried out in accordance with local regulations.

Before making any connections to the unit or the feed pump, switch off the power supply, and make sure that it cannot be accidentally switched on.

We recommend to fit an earth leakage circuit breaker (ELCB) to the incoming power supply.

Check that the RCME unit and feed pump are suitable for the power supply at the installation site.

All electrical connections must comply with enclosure class IP54.



Caution

8.1 Electrical connection

The following connections must be made before the system can be commissioned.

8.1.1 Connection of power supply

Check that the conductor cross-section of the power supply cable used is suitable for the maximum current (I_{max}) stated on the nameplate of the RCME unit.

Caution

Connect the power supply as shown in the wiring diagram supplied with the RCME control cabinet.

8.1.2 Connection of hydrostatic pressure sensor for collecting tank

Connect the supplied hydrostatic pressure sensor as shown in the wiring diagram supplied with the RCME control cabinet.

Enter the tank depth and desired low-level limit for the collecting tank in display 4.1. See section 15.1 *Tank settings (4.1)*.

8.1.3 Connection of feed pump

Connect the feed pump as shown in the wiring diagram supplied with the RCME control cabinet.

8.1.4 Connection of flow sensor

Connect the flow sensor as shown in the wiring diagram supplied with the RCME control cabinet.

8.2 Motor protection

8.2.1 CME 5-4 booster pump

The motor incorporates thermal protection against slow overloading, blocking and dry running.

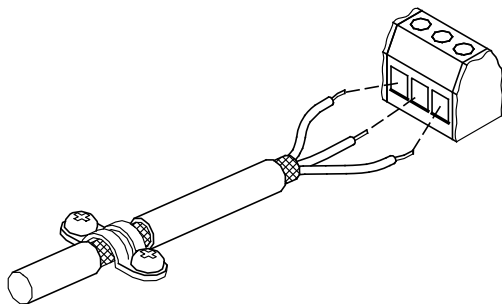
8.2.2 SPO 5-55 feed pump

The feed pump is protected by a motor-protective circuit breaker with automatic resetting. The motor-protective circuit breaker is located inside the control cabinet and has been set from factory.

8.3 EMC-correct installation

In order to ensure an EMC-correct installation, these guidelines for communication and signal cables must be followed:

- Leave the screen of the cable as close to the connecting terminals as possible.
- Fix the screen with a cable clamp.



TM02 1325 0901

Fig. 3 Example of stripped cable with screen

- Connect the screen of communication and signal cables to frame at both ends.
- Screws for frame connections must always be tightened whether a cable is fitted or not.

9. Control panel

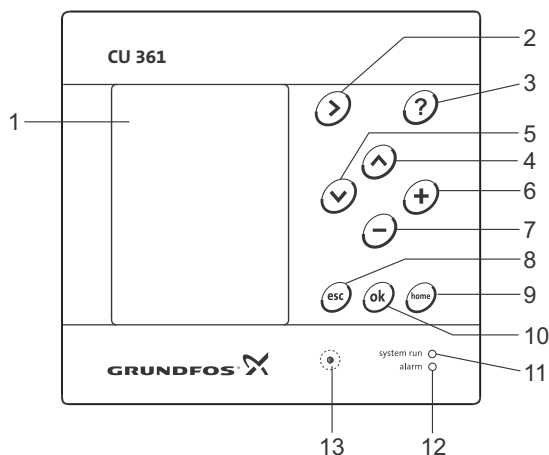


Fig. 4 CU 361 control panel

TM04 2394 2508

Pos.	Button/ indicator light	Description
1		LCD display
2		Changes to next column in menu structure Press [→] to go to the next column in the menu structure. If [→] is pressed in the last column, it changes to the first column to the left.
3		Changes to help text Press [?] to display the relevant help text. All the elements in the display have a selectable help text. Press [?] again to close the help text window.
4		Goes up in lists Press [↑] to go up in lists. If [↑] is pressed on the top line of the list, it moves to the bottom of the list.
5		Goes down in lists Press [↓] to go down in lists. If [↓] is pressed on the bottom line of the list, it moves to the top of the list.
6		Increases the value of a selected parameter Press [+] to increase the value of a selected parameter.
7		Decreases the value of a selected parameter Press [-] to decrease the value of a selected parameter.
8		Goes one display back Press [esc] to return to the previous display in the menu. Press [esc] to cancel a change.
9		Goes back to the "Status" menu Press [home] to return to the system overview display.
10		Saves a value Use [ok] as enter button. Also use [ok] to begin the process of changing a value and to save the change.
11		Green indicator light (operation) The green indicator light is on when the power supply has been switched on.
12		Red indicator light (alarm) The red indicator light is on if there is an alarm or a warning. The fault can be identified from the alarm list.
13		Changes the contrast of the display Press the contrast button to change the display contrast. Press [+] or [-] to increase or decrease the contrast.

Active buttons are backlit.

9.1 Display layout

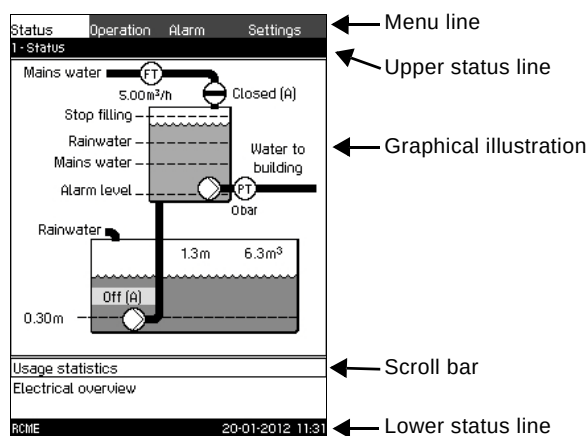


Fig. 5 Display layout

Menu line

The display has four main menus:

Status	Indication of system status
Operation	Changes of operating parameters (password option)
Alarm	Alarm log for fault finding purposes
Settings	Change of system settings (password option)

Upper status line

The upper status line shows the following:

- number and name of the display (to the left)
- the symbol  in case of alarm (to the right)
- the symbol  if the service language has been selected (to the right).

Lower status line

The lower status line shows the following:

- system name (to the left)
- date and time (to the right).

Graphical illustration

The graphical illustration shows the RCME rainwater system, including feed pump and collecting tank. The display also shows various settings as well as system status, a historical indication or other elements.

Scroll bar

If the list of elements exceeds the display, the symbols  and  will appear in the scroll bar to the right. Use  and  to move up and down in the list.

10. Commissioning

10.1 Checks before first start-up

Check the following before the system is started up for the first time:

- Mechanical installation has been made as described in section 7. *Mechanical installation*.
- Electrical installation has been made as described in section 8. *Electrical installation*.
- All isolating valves are open.
- The precharge pressure of the integrated pressure vessel is set to 70 % of the desired cut-in pressure. For further information, see section 17.3 *Integrated pressure vessel*.
- The collecting tank has been filled sufficiently with water in order for the SPO pump to deliver water to the buffer tank.

10.2 Start-up

1. Switch on the power supply, and wait for the first display to appear.
 2. Set "Display language", "Units", "Date and time" in displays 4.5.1 to 4.5.3. See sections 15.5.1 *Display language (4.5.1)* to 15.5.3 *Date and time (4.5.3)*.
 3. Make the collecting tank settings. See section 15.1 *Tank settings (4.1)*.
 4. Check the maximum permissible discharge pressure for the booster pump (p_{max}) and adjust, if necessary. See section 15.2 *Booster pump settings (4.2)*.
 5. Clear any alarms. See section 14.1 *Current alarms (3.1)*.
 6. Select and define the conditions to be monitored. See section 15.3.1 *Digital inputs (4.3.1)*.
 7. Ensure that the feed pump is connected electrically with the correct phase sequence so that the direction of rotation is correct. The feed pump can be started manually in order to check that the pump is connected correctly. See section 13. *Operation (2)*.
 8. Set the feed pump to "Auto". Set the mains water valve to "Auto". Set the booster pump to "Auto". See section 13. *Operation (2)*.
 9. Activate rainwater consuming equipment, for example toilets or garden hose, in order to prime the system and pipework.
- The RCME system is now ready to supply rainwater to the building.

11. Menus

Functions tree

1. Status (page 11)

- 1.1 Electrical overview (page 11)
 - 1.1.1 Analog inputs (page 11)
 - 1.1.2 Digital inputs (page 11)
 - 1.1.3 Digital outputs (page 12)
- 1.2 Usage statistics (page 12)

2. Operation (page 12)

3. Alarm (page 12)

- 3.1 Current alarms (page 13)
- 3.2 Alarm log (page 13)

4. Settings (page 14)

- 4.1 Tank settings (page 14)
- 4.2 Booster pump settings (page 15)
- 4.3 Inputs and outputs (page 15)
 - 4.3.1 Digital inputs (page 15)
 - 4.3.1.1 Digital inputs and functions (page 16)
 - 4.3.2 Digital inputs (page 16)
 - 4.3.2.1 Analog inputs (page 16)
 - 4.3.3 Digital outputs (page 17)
- 4.4 User-defined alarms (page 17)
- 4.5 Basic settings, CU 361 (page 17)
 - 4.5.1 Display language (page 18)
 - 4.5.2 Units (page 18)
 - 4.5.2.1 Pressure
 - 4.5.2.2 Flow rate
 - 4.5.2.3 Volume
 - 4.5.2.4 Level
 - 4.5.3 Date and time (page 18)
 - 4.5.4 Password (page 19)
 - 4.5.5 Ethernet (page 19)
 - 4.5.6 GENIbus number (page 19)
 - 4.5.7 Software status (page 19)

12. Status (1)

When the RCME control cabinet is switched on, this status display will appear. No settings can be made in this menu.

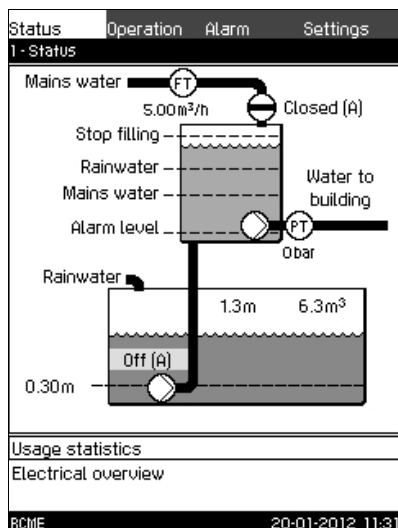


Fig. 6 Status

If the buttons on the control panel have not been touched for 15 minutes, the display will return to this display, and the back light of the panel will be dimmed.

12.1 Electrical overview (1.1)

This display shows an overview of the various inputs and outputs. The submenus in this display are described in the following sections.

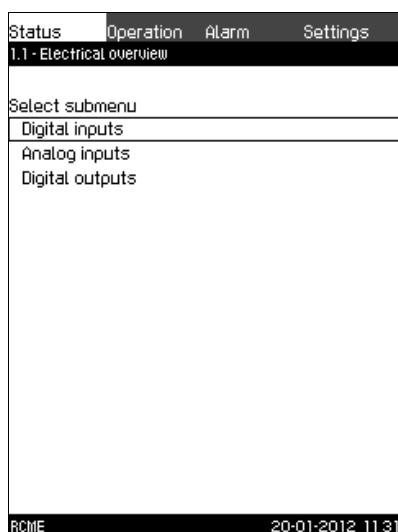


Fig. 7 Electrical overview

12.1.1 Analog inputs (1.1.1)

This display shows an overview of the individual analog inputs and the current measured values of each input.

The factory configuration is shown in fig. 8.

Example

The AI1 analog input on the CU 361 (designated terminal 51) is linked to the input "Level in collecting tank".

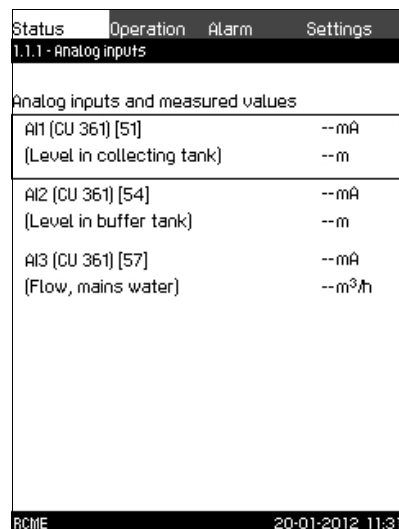


Fig. 8 Analog inputs

12.1.2 Digital inputs (1.1.2)

This display shows the status of the individual digital inputs.

The factory configuration is shown in fig. 9.

Example

The DI1 digital input on the CU 361 (designated terminal 10) is linked to the input "Fault signal from feed pump".

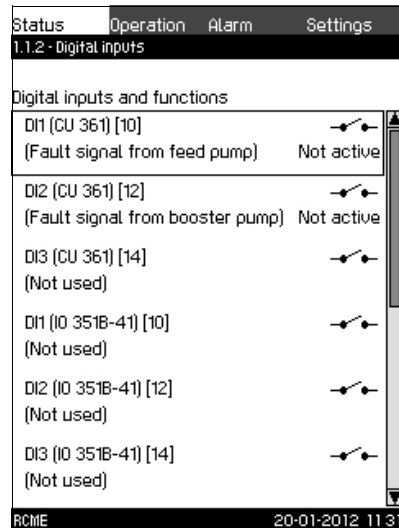


Fig. 9 Digital inputs

Display_1.1.1

Display_1.1.2

12.1.3 Digital outputs (1.1.3)

This display shows the status of the individual digital outputs. The factory configuration is shown in fig. 10.

Example

The DO1 digital output on the CU 361 (designated terminal 71) is linked to "Mains water valve open".

Status	Operation	Alarm	Settings
1.1.3 - Digital outputs			
Digital outputs and functions			
DO1 (CU 361) [71] (Mains water valve open)	Not active		
DO2 (CU 361) [74] (Feed pump running)	Not active		
DO1 (IO 351B-41) [77] (Feed pump alarm)	Not active		
DO2 (IO 351B-41) [79] (Booster pump alarm)	Not active		
DO3 (IO 351B-41) [81] (High-water alarm in buffer tank)	Not active		
DO4 (IO 351B-41) [82] (Mains water valve open)	Not active		
RCME 20-01-2012 11:31			

Fig. 10 Digital outputs

12.2 Usage statistics (1.2)

This display shows statistics for the mains water valve and booster pump.

Status	Operation	Alarm	Settings
1.2 - Usage statistics			
Last reset	01-01-2010 00:00		
Mains water valve			
Valve openings, total	50		
Valve openings, last hour	0		
Last opened	22-11-2011 08:20		
Valve open, total time	17:08:13		
Valve open, longest time	2:15:43		
Valve open, latest time	0:01:10		
Volume	10.67 m ³		
Booster pump			
Pump starts, total	50		
Pump starts, last hour	0		
Last started	22-11-2011 08:22		
Total operating time	0:45:18		
Longest operating time	0:03:40		
Latest operating time	0:01:38		
RCME 20-01-2012 11:31			

Fig. 11 Usage statistics

13. Operation (2)

In this display, it is possible to change the operating status for the mains water valve and the feed pump. Furthermore, it is possible to reset the usage statistics.

Note *In case of an interruption of the power supply, the system will be started with the last made settings.*

Status	Operation	Alarm	Settings
2 - Operation			
Mains water valve			
Auto		☒	
Open		☐	
Closed		☐	
Booster pump			
Auto		☒	
On		☐	
Off		☐	
Feed pump			
Auto		☒	
On		☐	
Off		☐	
Reset usage statistics			
RCME 20-01-2012 11:31			

Fig. 12 Operation

14. Alarm (3)

From this display, it is possible to access a list of current alarms and the alarm log.

Status	Operation	Alarm	Settings
3 - Alarm			
Further information about alarms, go to			
Current alarms			
Alarm log			
RCME 20-01-2012 11:31			

Fig. 13 Alarm

14.1 Current alarms (3.1)

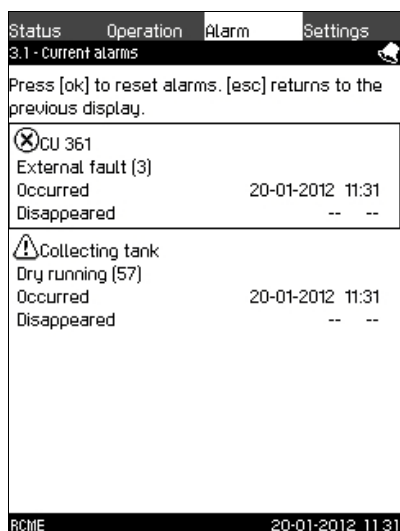


Fig. 14 Current alarms

This menu shows the following:

- Warnings caused by faults that still exist.
- Warnings caused by faults that have disappeared, but the warning requires manual resetting.
- Alarms caused by faults that still exist.
- Alarms caused by faults that have disappeared, but the alarm requires manual resetting.

All warnings and alarms with automatic resetting are automatically removed from the display when the fault has been corrected.

Alarms requiring manual resetting are reset in this display by pressing [ok]. An alarm cannot be reset until the fault has been corrected.

For every warning or alarm, the following is shown:

- Whether it is a warning or an alarm .
- Where the fault occurred: Feed pump alarm, Booster pump alarm, etc.
- In the case of input-related faults, the input will be shown.
- What the cause of the fault is, and the alarm code in brackets: Water shortage (214), Pressure above maximum pressure (210), etc.
- When the fault occurred: Date and time.
- When the fault disappeared: Date and time. If the fault still exists, date and time are shown as --:--:--.

The latest warning or alarm is shown at the top of the display.

14.2 Alarm log (3.2)



Fig. 15 Alarm log

This menu shows the following:

- Warnings caused by faults that still exist.
- Alarms caused by faults that still exist.
- Alarms caused by faults that have disappeared, but the alarm requires manual resetting.

All warnings and alarms with automatic resetting are automatically removed from the display when the fault has been corrected.

Alarms requiring manual resetting are reset in this display by pressing [ok]. An alarm cannot be reset until the fault has been corrected.

For every warning or alarm, the following is shown:

- Whether it is a warning or an alarm .
- Where the fault occurred: Sensor type (flow rate, pressure, etc.), external fault, communication fault, etc.
- In the case of input-related faults, the input will be shown.
- What the cause of the fault is, and the alarm code in brackets. This is very important to enable the Grundfos service engineer to identify and correct the fault.
- When the fault occurred: Date and time.
- When the fault disappeared: Date and time. If the fault still exists, date and time are shown as --:--:--.

The latest warning or alarm is shown at the top of the display.

Display_3.1

Display_3.2

15. Settings (4)

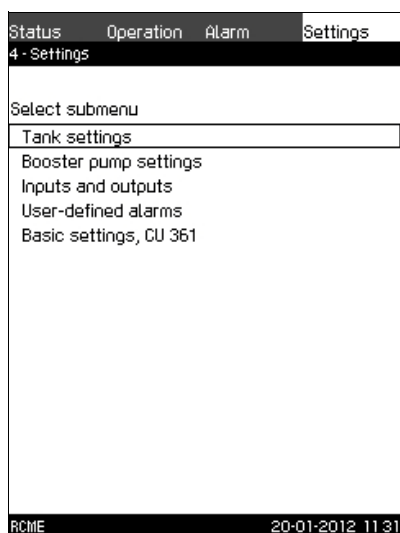


Fig. 16 Settings

In this menu, it is possible to access the following submenus in order to change settings.

Submenus:

- Tank settings
- Booster pump settings
- Inputs and outputs
- User-defined alarms
- Basic settings, CU 361.

15.1 Tank settings (4.1)

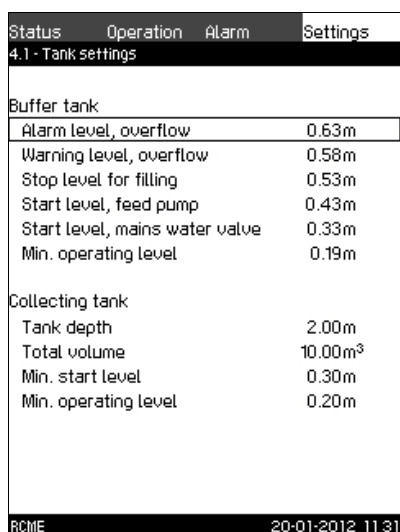


Fig. 17 Tank settings

In this menu, settings can be made for the buffer tank and the collecting tank.

15.1.1 Buffer tank settings

Note *The levels for the buffer tank have been set correctly from the factory and should not be changed.*

- Alarm level, overflow
At this level, an alarm for overflow is activated. The level is factory-set and cannot be changed.
- Warning level, overflow
At this level, a warning for overflow is activated as there is a risk of overflow. The level is factory-set and cannot be changed.
- Stop level for filling
At this level, the filling of the buffer tank must stop. The tank is filled either by the feed pump or via the mains water valve. The level is factory-set and cannot be changed.
- Start level, feed pump
At this level, the feed pump starts supplying rainwater to the buffer tank. The level can be changed.
- Start level, mains water valve
At this level, the mains water valve opens, and mains water is supplied to the buffer tank. The level can be changed.
- Min. operating level
At this level, the booster pump is no longer allowed to run and is stopped. The level is factory-set and cannot be changed.

Factory settings

Level	Setting
Alarm level, overflow (fixed value)	0.63 m
Warning level, overflow (fixed value)	0.58 m
Stop level for filling (fixed value)	0.53 m
Start level, feed pump	0.43 m
Start level, mains water valve	0.33 m
Min. operating level (fixed value)	0.19 m

15.1.2 Collecting tank settings

- Tank depth
Enter the depth of the collecting tank.
- Total volume
Enter the total volume of the collecting tank.
- Min. start level
Enter the minimum level at which the feed pump is allowed to start.
- Min. operating level
Enter the minimum operating level at which the feed pump is no longer allowed to run and will be stopped.

Factory settings

Level	Setting
Tank depth	2.00 m
Total volume	10.00 m ³
Min. start level	0.30 m
Min. operating level	0.20 m

Display_4

Display_4.1

15.2 Booster pump settings (4.2)

Status	Operation	Alarm	Settings
4.2 - Booster pump settings			
Booster pump			
Setpoint			4.5bar
Max. pressure pmax			5bar
RCME			20-01-2012 11:31

Fig. 18 Booster pump settings

In this menu, the booster pump setpoint is set and the maximum allowed discharge pressure ($p_{\max}$) can be set with 1 bar increments. The chosen setting prevents the setpoint from being set to a value higher than $p_{\max}$.

Factory setting

Pressure	Setting
Setpoint	4.5 bar
$p_{\max}$	5 bar

15.3 Inputs and outputs (4.3)

Status	Operation	Alarm	Settings
4.3 - Inputs and outputs			
Select submenu			
Digital inputs			
Analog inputs			
Digital outputs			
Modules installed			
IO 351B-41			☒
RCME 20-01-2012 11:31			

Fig. 19 Inputs and outputs

In this menu, the various analog and digital signal transmitters are configured.

This menu has these submenus:

- Digital inputs
- Analog inputs
- Digital outputs.

15.3.1 Digital inputs (4.3.1)

As standard, the CU 361 has three digital inputs. Furthermore the RCME incorporates an IO 351B module with nine digital inputs.

In this menu, the digital inputs of the CU 361 and IO 351B can be set. The inputs DI1, DI2 and DI3 of the CU 361 have been factory-configured as have DI1 of the IO 351B module.

The remaining eight inputs can be defined by the user.

In the display, all digital inputs are shown so that their physical position in the control cabinet can be identified.

Example

DI1 (IO 351B-41) [10]:

DI1	Digital input 1
IO 351B-41	IO 351B, GENIbus number 41
[10]	Terminal 10

For further information on the connection of the various digital inputs, see the wiring diagram supplied with the control cabinet.

This display shows the three digital inputs of the CU 361 and the nine digital inputs of the IO 351B.

Status	Operation	Alarm	Settings
4.3.1 - Digital inputs			
Digital inputs and functions			
DI1 (CU 361) [10]	(Fault signal from feed pump)		
DI2 (CU 361) [12]	(Fault signal from booster pump)		
DI3 (CU 361) [14]	(Not used)		
DI1 (IO 351B-41) [10]	(Not used)		
DI2 (IO 351B-41) [12]	(Not used)		
DI3 (IO 351B-41) [14]	(Not used)		
DI4 (IO 351B-41) [16]	(Not used)		
DI5 (IO 351B-41) [20]	(Not used)		
RCME		20-01-2012 11:31	

Fig. 20 Digital inputs

Standard configuration:

DI1 (CU 361) [10]	Fault signal from feed pump
DI2 (CU 361) [12]	Fault signal from booster pump
DI3 (CU 361) [14]	Not used
DI1 (IO 351B-41) [10]	Not used
DI2 (IO 351B-41) [12]	Not used
DI3 (IO 351B-41) [14]	Not used
DI4 (IO 351B-41) [16]	Not used
DI5 (IO 351B-41) [20]	Not used
DI6 (IO 351B-41) [24]	Not used
DI7 (IO 351B-41) [42]	Not used
DI8 (IO 351B-41) [44]	Not used
DI9 (IO 351B-41) [46]	Not used

15.3.2 Digital inputs and functions (4.3.1.1)

Status	Operation	Alarm	Settings
4.3.1.1 - Digital inputs and functions			
Select input logic			
NO (normally open)			☒
NC (normally closed)			☐
Function, DI1 (CU 361)			
Not used			☐
External fault			☐
Resetting of alarm			☐
Fault signal from feed pump			☒
Fault signal from booster pump			☐
Feed pump disabled			☐
Mains water valve disabled			☐
RCME 20-01-2012 11:31			

Display_4.3.1.1

Fig. 21 Digital inputs and functions

In this display a function can be related to the digital inputs.

Setting range

Input logic
NO (normally open)
NC (normally closed)
Functions
Not used
External fault
Resetting of alarm
Fault signal from feed pump
Fault signal from booster pump
Feed pump disabled
Mains water valve disabled

15.3.3 Analog inputs (4.3.2)

Status	Operation	Alarm	Settings
4.3.2 - Analog inputs			
Analog inputs and measured values			
AI1 (CU 361) [51]		(Level in collecting tank)	
AI2 (CU 361) [54]		(Level in buffer tank)	
AI3 (CU 361) [57]		(Flow, mains water)	
AI1 (IO 351B-41) [57]		(Not used)	
AI2 (IO 351B-41) [60]		(Not used)	
AI1 (IO 351B-42) [57]		(Not used)	
AI2 (IO 351B-42) [60]		(Not used)	
RCME 20-01-2012 11:31			

Display_4.3.2

Fig. 22 Analog inputs

This display shows the analog inputs of the RCME unit. Only the following sensors can be connected to the system:

- Level in collecting tank
- Level in buffer tank
- Flow, mains water.

Standard configuration:

AI1 (CU 361) [51]	Level in collecting tank
AI2 (CU 361) [54]	Level in buffer tank
AI3 (CU 361) [57]	Flow, mains water
AI1 (IO 351B-41) [57]	Not used
AI2 (IO 351B-41) [60]	Not used
AI1 (IO 351B-42) [57]	Not used
AI2 (IO 351B-42) [60]	Not used

15.3.4 Analog inputs (4.3.2.1 - 4.3.2.3)

Configure the analog inputs in these displays.

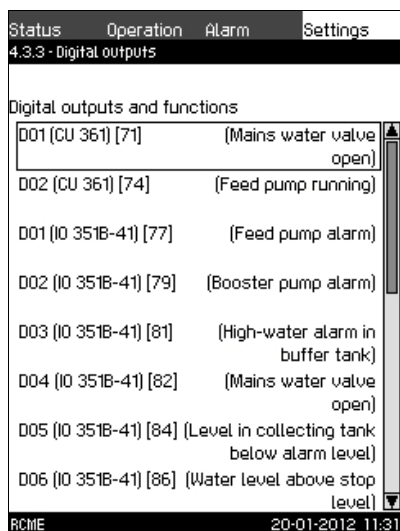
The display below shows an analog input which has been configured for the measurement of the level in the collecting tank.

Status	Operation	Alarm	Settings
4.3.2.1 - Analog inputs			
Setting, AI1 (CU 361) [51]			
0-20 mA			☐
4-20 mA			☒
0-10 V			☐
Not used			☐
Measured input value			
(Level in collecting tank)			
Range			
Min.	0.00m		
Max.	6.00m		
RCME 20-01-2012 11:31			

Display_4.3.2.1

Fig. 23 Analog inputs

15.3.5 Digital outputs (4.3.3)



Display_4.3.3

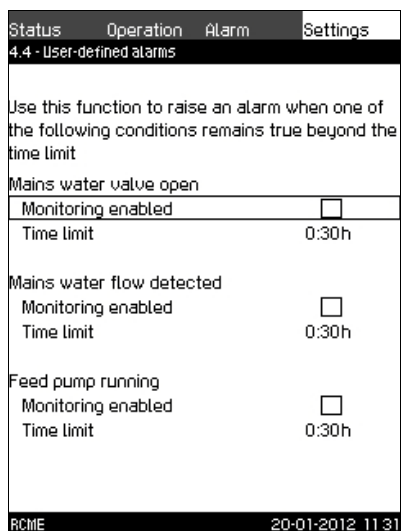
Fig. 24 Digital outputs

This display shows the two digital outputs of the CU 361 and the seven digital outputs of the IO 351B.

Standard configuration:

DO1 (CU 361) [71]	Mains water valve open
DO2 (CU 361) [74]	Feed pump running
DO1 (IO 351B-41) [77]	Feed pump alarm
DO2 (IO 351B-41) [79]	Booster pump alarm
DO3 (IO 351B-41) [81]	High-water alarm in buffer tank
DO4 (IO 351B-41) [82]	Mains water valve open
DO5 (IO 351B-41) [84]	Level in collecting tank below alarm level
DO6 (IO 351B-41) [86]	Water level above stop level
DO7 (IO 351B-41) [88]	No function

15.4 User-defined alarms (4.4)



Display_4.4

Fig. 25 User-defined alarms

In this menu, it is possible to enable monitoring of the following conditions:

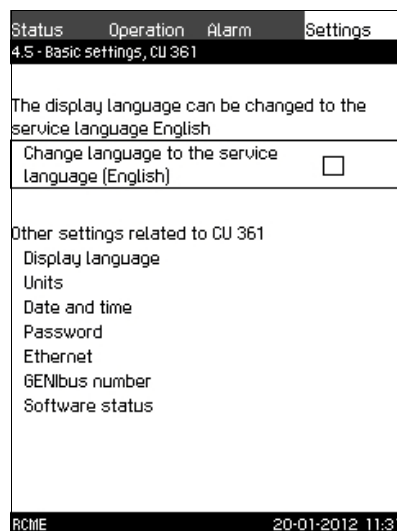
- Mains water valve open
- Mains water flow detected
- Feed pump running.

For each of the monitored conditions, a time limit can be set. The time limit determines the delay between detection of the condition and activation of the alarm.

15.5 Basic settings, CU 361 (4.5)

In this menu, the basic settings of the CU 361 can be made:

- Activation of service language, English
- Other settings related to CU 361
 - Display language
 - Units
 - Date and time
 - Password for "Operation" and "Settings" menus
 - Ethernet
 - GENIbus number
 - Software status.



Display_4.5

Fig. 26 Basic settings, CU 361

Note

If the service language is selected, the symbol  is shown in the upper right corner of all displays.

If the buttons are not touched for 15 minutes, the display will return to the language selected at start-up or to the language selected in display 4.5.1.

15.5.1 Display language (4.5.1)

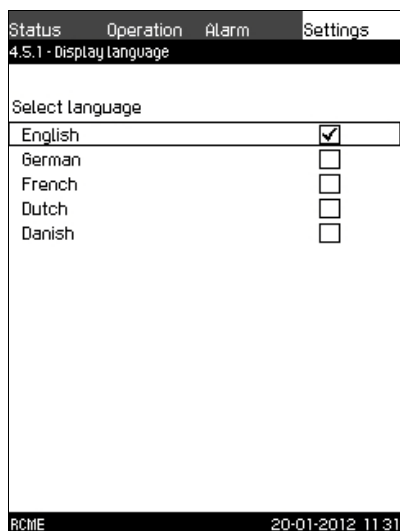


Fig. 27 Display language

Select the display language of the CU 361.

Possible settings:

- English
- German
- French
- Dutch
- Danish.

Factory setting

The display language is English. It can be changed at start-up.

15.5.2 Units (4.5.2)

Select measuring units for the various parameters shown in the display.

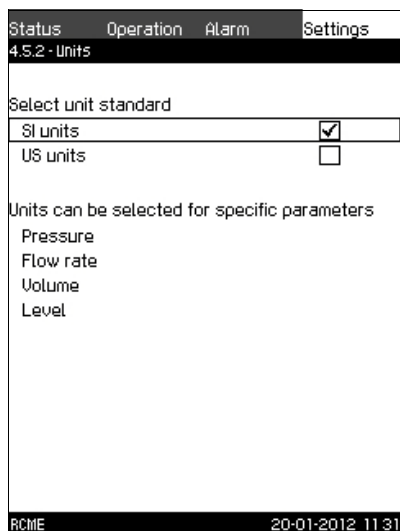


Fig. 28 Units

For the basic setting, choose between SI and US units. It is also possible to select other units for the individual parameters.

Setting range

Parameter	Basic setting		Possible units
	SI	US	
Pressure	bar	psi	kPa, MPa, mbar, bar, m, psi
Flow rate	m ³ /h	gpm	m ³ /s, m ³ /h, l/s, gpm, yd ³ /s, yd ³ /min, yd ³ /h
Volume	m ³	gal	l, m ³ , gal, yd ³
Level	m	ft	m, cm, ft, in

Note If units are changed from SI to US and vice versa, all individually set parameters are changed to the basic setting in question.

Factory setting

The CU 361 has been factory-set to SI units.

15.5.3 Date and time (4.5.3)

Set the date and time and select how they are to be shown in the display.

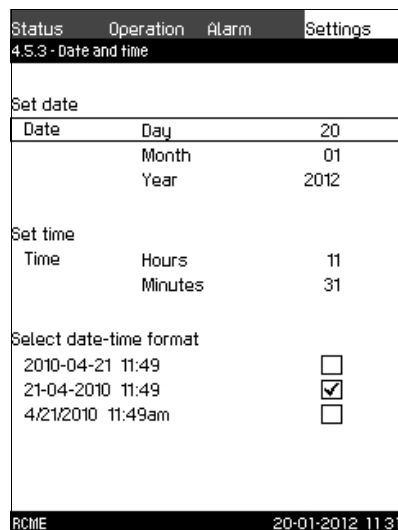


Fig. 29 Date and time

The clock has a built-in rechargeable power supply which can supply the clock for up to 20 days if the power supply to the CU 361 is interrupted.

If the clock has been without power for more than 20 days, it must be set again.

Setting range

The date can be set as day, month and year. The time can be set as a 24-hour clock showing hours and minutes.

There are three formats:

Examples of format

2010-04-21 11:49
21-04-2010 11:49
4/21/2010 11:49 am

Factory setting

Local time.

Note If the CU 361 has been without power for more than 20 days since it left the factory, the clock may have returned to the original setting: 01-01-2008 0:00.

Date and time may have been changed during the setting of the CU 361.

There is no automatic changeover to and from daylight-saving time.

15.5.4 Password (4.5.4)

In this display, it is possible to limit the access to the menus "Operation" and "Settings" by means of a password.

Status	Operation	Alarm	Settings
4.5.4 - Password			
A 4-digit password can be enabled for the Operation and Settings menus			
Password, Operation menu			
Password enabled ☐			
Enter password 1 2 3 4			
Password, Settings menu			
Password enabled ☐			
Enter password 1 2 3 4			
RCME 20-01-2012 11:31			

Display_4.5.4

Fig. 30 Password

If the access is limited, it will not be possible to view or set any parameters in the menus.

The password must consist of four digits and may be used for both menus.

Note If you have forgotten your password, contact Grundfos.

Setting via control panel

1. Select the password to be enabled and press [ok].
2. Select "Enter password" and press [ok].
Now the first digit of the password is flashing.
3. Select digit with [+] and [-] and save with [ok].
Now the second digit of the password is flashing.
4. Repeat points 1 to 3 if the another password is to be enabled.

Factory setting

Both passwords are disabled. If a password is to be enabled, the factory setting will be "1234".

15.5.5 Ethernet (4.5.5)

Status	Operation	Alarm	Settings
4.5.5 - Ethernet			
Host name rcme			
Use DHCP ☒			
From DHCP server			
IP address	0	0	0 0
Subnet mask	0	0	0 0
Standard gateway	0	0	0 0
MAC address	00 14 44 00 00 00		
Press [ok] to reset the Ethernet password			
RCME 20-01-2012 11:31			

Display_4.5.5

Fig. 31 Ethernet

The CU 361 has an Ethernet connection for communication with a computer, either directly or via internet. For further information, see section 16.1 Ethernet.

15.5.6 GENIbus number (4.5.6)

In this display, a GENIbus number for external communication can be set.

Status	Operation	Alarm	Settings
4.5.6 - GENIbus number			
GENIbus number for external communication			
Number 1			
RCME 20-01-2012 11:31			

Display_4.5.6

Fig. 32 GENIbus number

The RCME system can be connected to a network and become part of a superior building management system. For further information, see fig. 34 and section 16.2 GENIbus.

The communication is carried out according to the Grundfos bus protocol, GENIbus, and enables connection to a building management system or another external control system.

Operating parameters can be set via the bus signal. Furthermore, status of important parameters and fault indications can be read from the CU 361.

For further information, contact Grundfos.

Setting range

The number can be set between 1 and 64.

Setting via control panel

1. Select "GENIbus number" with [+] and [-] and press [ok].
2. Select number with [+] and [-] and save with [ok].

Factory setting

41.

15.5.7 Software status (4.5.7)

Status	Operation	Alarm	Settings
4.5.7 - Software status			
Software versions			
Software version, CU 361 v01.01.01			
Press [ok] for software update			
Software update state			
Idle			
RCME 20-01-2012 11:31			

Display_4.5.7

Fig. 33 Software status

This display shows the version of the software installed in the CU 361.

As it is a status display, no settings can be made.

16. Data communication

The CU 361 incorporates hardware that enables communication with external units, such as a computer, via an external GENIbus or Ethernet connection.

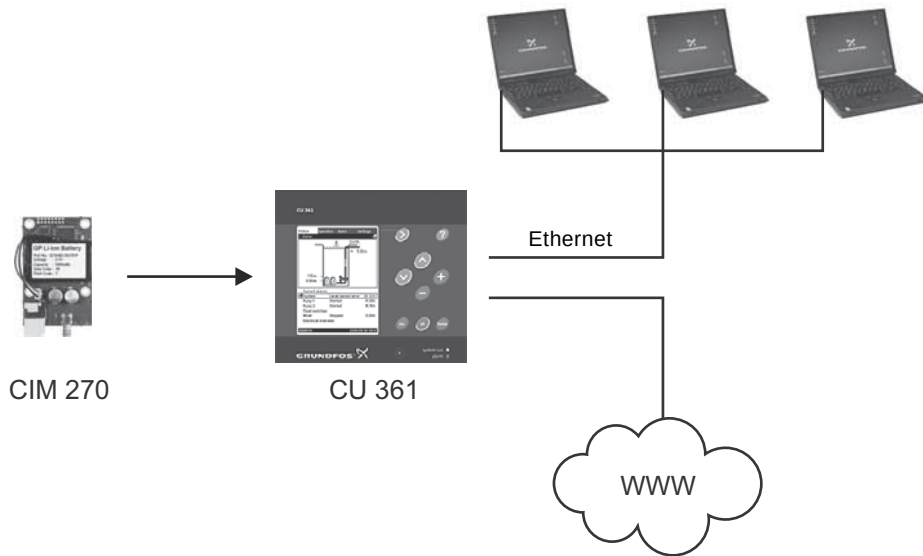


Fig. 34 Data communication via external GENIbus and Ethernet connections

16.1 Ethernet

Ethernet is the most widely used standard for local networks (LAN). The standardisation of this technology has created some of the easiest and cheapest ways of creating communication between electrical units, for instance between computers or between computers and control units.

The web server of the CU 361 makes it possible to connect a computer to the CU 361 via an Ethernet connection. The user interface can thus be exported from the CU 361 to a computer so that the CU 361 and consequently the RCME system can be remotely monitored and controlled.

Note

We recommend that you protect the connection to the CU 361 in consultation with the system administrator according to your safety requirements.

In order to use the web server, you must know the IP address of the CU 361. All network units must have a unique IP address to communicate with each other.

It is possible to change the factory-set IP address by configuration or by activating a DHCP (Dynamic Host Configuration Protocol), either directly in the CU 361 or via the web server. See the example in fig. 35.

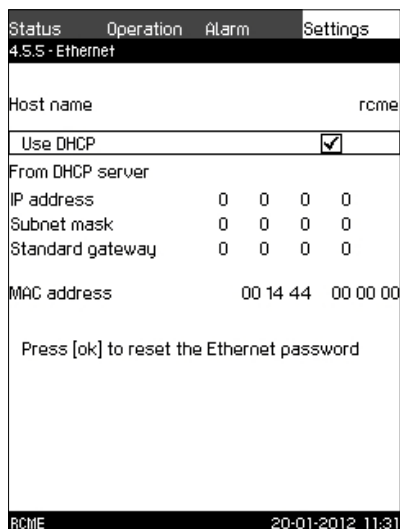


Fig. 35 Example of Ethernet setting

Dynamic assignment of an IP address for the CU 361 requires a DHCP server in the network. The DHCP server assigns a number of IP addresses to the electrical units and makes sure that two units do not receive the same IP address.

Use a standard internet browser when connecting to the web server of the CU 361.

If you want to use the factory-set IP address, no changes are required in the display. Open the internet browser, and enter the IP address of the CU 361.

In order to use dynamic assignment, the function must be enabled. Click [ok] "Use DHCP". A tick shows that the function has been enabled. Open the internet browser, and enter the CU 361 "Host name" instead of the IP address (the host name is stated at the top of display 4.5.5). The internet browser will now try to establish connection to the CU 361.

DHCP requires a host name.

Note

The host name can only be changed by means of a GSC file (configuration file) or via a web server. See section Change of network setting, page 21.

This is the first display that appears when connecting to the CU 361.

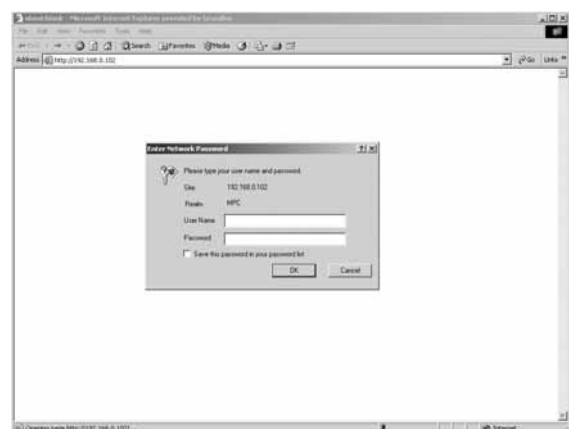


Fig. 36 Connecting to the CU 361

Factory setting

User name: admin

Password: admin

When user name and password have been entered, a Java Runtime Environment application starts up in the CU 361, provided that it has been installed on the computer in question. If this is not the case, but the computer is connected to the internet, then use the link on the screen to download and install the Java Runtime Environment application.



Fig. 37 Display with link to Java Runtime Environment

The Java Runtime Environment application transfers the CU 361 user interface (including display and control panel) to the computer screen. It is now possible to monitor and control the CU 361 from the computer.



Fig. 38 Network setting

Change of network setting

When the connection to the web server of the CU 361 has been established, it is possible to change the network setting.



Fig. 39 Change of network setting

1. Click the icon "Network admin".
2. Enter the changes.
3. Click [Submit] to enable the changes.

Change of password



Fig. 40 Change of password

1. Click the icon "Change password".
2. Enter the new password.
3. Click [Submit] to enable the new password.

16.2 GENIBus

When a GENIBus module is installed, it is possible to connect a CU 361 to an external network. The connection can be established via a GENIBus-based network or a network based on another protocol via a gateway. See fig. 34. For further information, contact Grundfos.

TM03 2050 3505

TM03 2049 3505

TM04 9436 4310

TM03 2051 3505

17. Maintenance



Warning

Before starting any work on pumps, terminal boxes or control cabinet, the power supply must have been switched off for at least five minutes, and it must be ensured that it cannot be accidentally switched on.

17.1 CU 361

The CU 361 is maintenance-free. It must be kept clean and dry. Protect it from direct sunlight. The temperature of the CU 361 must not exceed the limits of ambient temperature. See section 19. *Technical data*.

17.2 Regular inspections

The RCME system should be inspected at regular intervals. This must be carried out when the system has been shut down.

Check the following:

- Check that there are no leaks on the internal pipework, the pump and the buffer tank.
- Check the precharge pressure of the pressure vessel and adjust, if necessary. See section 17.3 *Integrated pressure vessel*.
- Check that the booster pump operates quietly and smoothly without vibration or excessive noise.

Should any faults be found, see section 21. *Fault finding*.

After inspection ensure that the isolating valves are fully open, and that the power supply to the RCME system is switched on again.

17.3 Integrated pressure vessel

Set the precharge pressure of the vessel to 70 % of the setpoint selected on the booster pump control panel (booster pump discharge pressure).

Note

Measure the precharge pressure while the system is pressureless.

If the precharge pressure is not set correctly, this may result in incorrect operation of the RCME unit.

Check and adjust the precharge pressure of the vessel before the booster pump is filled with water. If the system has already been filled and the pump has been running, proceed as follows:

1. Switch off the power supply to the RCME system.
2. Release the water pressure by opening a tap in the discharge pipe of the system. Leave the tap open while checking and adjusting the precharge pressure.
3. The precharge pressure can now be checked using a car tyre pressure gauge. The air valve is located in the top right-hand corner of the pump mounting plate.
For small pressure adjustments a foot pump can be used. If large amounts of gas are required, use dry nitrogen to prevent corrosion within the vessel.

Setting example:

If the discharge pressure has been set to 4.5 bar, the precharge pressure must be $4.5 \times 0.7 = 3.15$ bar.

18. Taking the RCME system out of operation

Take the RCME system out of operation by switching the main switch off.



Warning

The conductors to the main switch are still live. Lock the main switch to ensure that the power supply cannot be accidentally switched on.

19. Technical data

19.1 Pressure

Maximum operating pressure: 5 bar.

19.2 Temperatures

Liquid temperature: +3 °C to +40 °C.

Maximum ambient temperature: +40 °C.

19.3 Relative air humidity

Maximum 95 %.

19.4 Sound pressure level

65 dB(A).

19.5 Enclosure class

IP54.

19.6 Volume of buffer tank

Rated volume: 160 litres.

Effective water volume: 108 litres.

Minimum booster pump operating volume: 41 litres.

19.7 Weight of RCME unit

Weight with filled buffer tank: 200 kg.

The stated weight includes base plate, booster pump and buffer tank.

19.8 Dimensions

See fig. 2 on page 103.

20. Electrical data

20.1 Supply voltage

3 x 400 VAC - 10 %/+ 10 %, 50 Hz, PE.

21. Fault finding



Warning

Before starting fault finding, switch off the power supply. Make sure that the power supply cannot be accidentally switched on.

Fault	Cause	Remedy
1. No rainwater is supplied to the buffer tank even though the collecting tank contains rainwater.	a) The feed pump is malfunctioning.	See installation and operating instructions for the feed pump.
	b) The feed pump has not been set to "Auto".	Set the feed pump to "Auto". See section 13. <i>Operation (2)</i> .
	c) The feed pump discharge pipe is blocked or cracked.	Remove blockage, or replace the pipe.
	d) Level sensor in collecting tank malfunctioning.	Check input from collecting tank level sensor on the status display. See section 12. <i>Status (1)</i> . Check sensor connections and sensor operation. Replace the sensor, if necessary.
	e) Level sensor in buffer tank malfunctioning.	Check input from buffer tank level sensor on the status display. See section 12. <i>Status (1)</i> . Check sensor connections and sensor operation. Replace the sensor, if necessary.
	f) Overload trip operated.	
2. The booster pump is running, but the discharge pressure is low.	a) Air in the booster pump.	Take the system out of operation by switching the main switch off. Slacken the vent screw on the back of the pump sleeve and wait until airless water comes out. Retighten the screw.
3. No rainwater is available, but mains water is not being supplied to the buffer tank.	a) The mains water valve has not been set to "Auto".	Set the mains water valve to "Auto". See section 13. <i>Operation (2)</i> .
	b) The mains water valve is malfunctioning.	Check whether the valve is blocked by impurities, and clean it, if necessary. Replace the valve, if necessary.
	c) User-defined alarm is activated.	Reset the alarm. See sections 15.4 <i>User-defined alarms (4.4)</i> and 14.1 <i>Current alarms (3.1)</i> .
4. The booster pump is unable to reach the setpoint.	a) The consumption is too high.	Reduce the consumption, if possible.
	b) Pipe fracture or leakage in the system.	Check the system, and repair, if necessary.
5. No connection to Ethernet.	a) Wrong host name.	Check the host name. See section 16.1 <i>Ethernet</i> .

22. Disposal

This product or parts of it must be disposed of in an environmentally sound way:

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.

Subject to alterations.

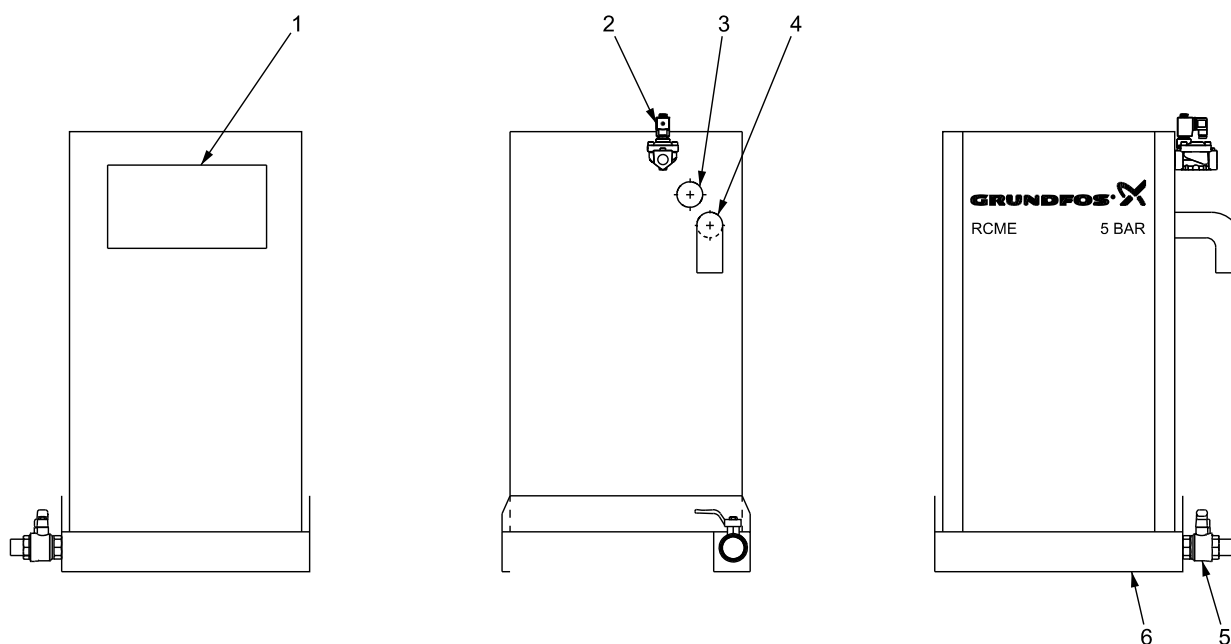


Fig. 1 RCME

1	GB	Cut-out for overflow purpose in order to comply with EN 1717.
	DK	Udslagsfelt til overløb for at kunne overholde EN 1717.
	DE	Rechteckiger Ausschnitt als Überlauf gemäß EN 1717.
	FR	Arrêt pour raison de trop-plein conformément à la norme EN 1717.
	NL	Afschakeling voor overstroomdoeleinden om te voldoen aan EN1717.
2	GB	Mains water valve
	DK	Ventil, vandværksvand
	DE	Leitungswasserzulaufventil
	FR	Vanne réseau
3	GB	Rainwater inlet
	DK	Tilgang for regnvand
	DE	Regenwasserzulauf
	FR	Entrée d'eau de pluie
4	GB	Overflow
	DK	Overløb
	DE	Überlauf
	FR	Trop plein
5	GB	Outlet connection
	DK	Afgangstilslutning
	DE	Abgang
	FR	Raccord de sortie
6	GB	Base plate
	DK	Fodplade
	DE	Grundplatte
	FR	Châssis
	NL	Uitlaataansluiting
	NL	Grondplaat

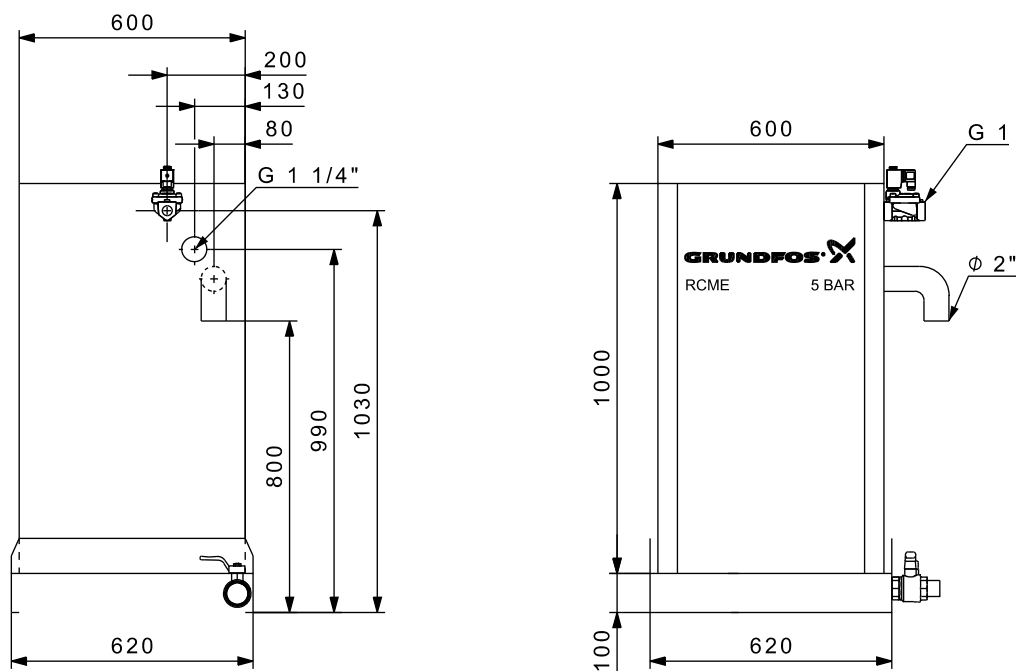


Fig. 2 Dimensional sketch

Dimensions are in mm.

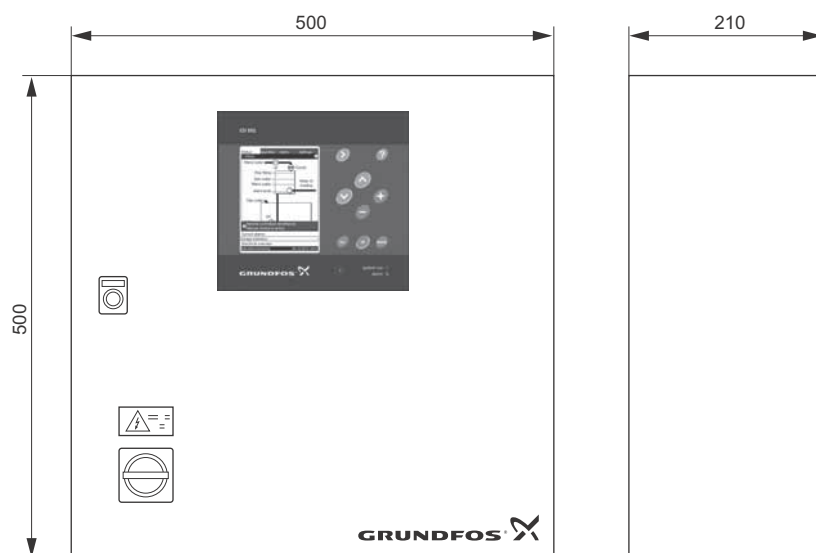


Fig. 3 Dimensional sketch of control cabinet

Dimensions are in mm.

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